2024-25

S6 MATH EP

M1

CUHKFAA CHAN CHUN HA SECONDARY SCHOOL

2024-2025 Secondary Six Final Examination

MATHEMATICS Extended Part

Module 1 (Calculus and Statistics)

Question-Answer Book

12/02/2025

8:15 am – 10:45 am (2½ hours)

This paper must be answered in English

Name: _____

Class: _____( )

	Marker Use Only
Question	Mark
1	
2	
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11	
12	

Total			

INSTRUCTIONS

1. After the announcement of the start of the examination, you should first write your Name, Class and Class number in the spaces provided on Page 1.

2. Attempt **ALL** questions in this paper. Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.

3. Graph paper and supplementary answer sheets will be supplied on request. Write your Name, Class and Class number and mark the question number box on each sheet, and fasten them with string **INSIDE** this book.

4. Unless otherwise specified, all working must be clearly shown.

5. Unless otherwise specified, numerical answers should be either exact or given to 4 decimal places.

6. For definite integrals, answers obtained by using numerical integration functions in calculators are not accepted.

7. The diagrams in this paper are not necessarily drawn to scale.

8. No extra time will be given to candidates for writing your Name, Class and Class number or filling in the question number boxes after 'Time is up' announcement.

Section A (50 marks)

1. The table below shows the probability distribution of a discrete random variable X, where a and b are constants.

x	0	1	2	3
$P(X = x)$	0.1	a	b	0.15

It is given that $\text{Var}(X) = 0.75$.

(a) Find the values of a and b.

(b) Let F be the event that $X \geq 1$ and G be the event that $X \leq 2$. Find $P(G \mid F)$.

(6 marks)

Answers written in the margins will not be marked.

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2. Let A and B be two events. Denote the complementary event of B by B'. It is given that

$P(A) = 2\,P(B')$, $P(A \mid B') - P(B' \mid A) = \frac{5}{13}$ and $P(A \cup B') = 0.58$.

(a) Find $P(A \mid B')$.

(b) Find $P(A)$ and $P(A \cap B)$.

(c) Are A and B independent? Explain your answer.

(7 marks)

Answers written in the margins will not be marked.

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3. In a factory, the manager inspects a batch of LED bulbs by randomly selecting 5 bulbs from the batch. If all the 5 LED bulbs are not defective, the batch will be accepted. According to the past record, 3% of the LED bulbs in a batch are defective.

(a) Find the probability that a batch of LED bulbs will not be accepted by the manager.

(b) Find the expected number of accepted batches of LED bulbs between two unaccepted batches.

(c) On average, the factory can produce 15 batches of LED bulbs in a week. A retailer has ordered 13 batches of LED bulbs in a certain week. Find the probability that the factory can complete the order in that week, with 13 accepted batches of LED bulbs deliver to the retailer. (6 marks)

Answers written in the margins will not be marked.

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4. The number of visitors to a theme park follows a Poisson distribution with a mean of 10 per minute. Let X be the number of visitors to the theme park in an hour.

(a) Write down the values of E(X) and Var(X).

(b) Let $\bar{X}$ be the mean of n independent random observations of X. Find the least value of n such that the estimated value of $P(590 \leq \bar{X} \leq 610)$ using central limit theorem is greater than 0.994.

(5 marks)

Answers written in the margins will not be marked.

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5. (a) Let $f(x) = e^{x^m} x^{m-1}$, where $m \neq 0$ and $m \neq 1$. Find $\int f(x)\,dx$ in terms of m.

(b) It is given that the region bounded by the curve $y = e^{x^k} x^{k-1}$, the x-axis and the straight lines $x = 1$ and $x = 8$ has an area of $\frac{e}{k}(e - 1)$, where $k \neq 0$ and $k \neq 1$. Find k.

(6 marks)

Answers written in the margins will not be marked.

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6. Let a be a constant.

(a) Expand e^{-ax} in ascending powers of x as far as the term in x^3.

(b) Let $y = (1 + ax)^5 e^{-ax}$. It is given that the coefficient of x^3 in the expansion of y is 63.

(i) Find a.

(ii) Someone claims that the graph of $y = (1 + ax)^5 e^{-ax}$ has only one turning point. Do you agree? Explain your answer.

(7 marks)

Answers written in the margins will not be marked.

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7. Let $y = \frac{x^3 - 4x}{\sqrt{2x^2 + 4}}$, where $-2 \leq x \leq 4$.

(a) Find $\frac{dy}{dx}$.

(b) Find the greatest value and the least value of y.

(6 marks)

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8. (a) By considering $\frac{d}{dx}[(2x^2+3x)e^{-2x}]$, find $\int(2x^2+x)e^{-2x}dx$.

(b) Define $f(x)=(4x^2+2x+1)e^{-2x}$ for all real numbers x. Let α and β be the two roots of the equation $f'(x)=0$, where $\alpha>\beta$. Find α and β. Also express $\int_{\beta}^{\alpha} f(x)\,dx$ in terms of e.

(7 marks)

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Section B (50 marks)

9. A company provides household appliances repair service. The number of requests received by the company for the repair service in a day follows a Poisson distribution with a mean of 2.1.

   (a) Find the probability that the company receives fewer than 6 requests for the repair service in a certain day. (2 marks)

   (b) The number of repair service requests received by the company on n randomly selected days are recorded. Find the least integral value of n such that the probability of having at least 1 day with more than 2 repair service requests is higher than 0.99. (2 marks)

   (c) It is known that the probability for a request to be charged \$1 000 or above is 0.6. The company will consider a day as a bad day if it receives fewer than 6 repair service requests and at most 1 of them is charged \$1 000 or above.

      (i) Find the probability that the company receives exactly 5 requests for the repair service on a certain day and that day is considered as a bad day.

      (ii) Find the probability that a certain Monday is considered as a bad day.

      (iii) Given that every day in a certain week is considered as a bad day, find the probability that at least 6 repair service requests received by the company in that week are charged \$1 000 or above.

      (9 marks)

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10. A wholesaler buys a large batch of brand *A* peaches. The weights of the batch of peaches follow a normal distribution with a mean of 284 g and a standard deviation of 20 g.

(a) (i) Find the probability that the weight of a randomly selected brand *A* peach lies between 274 g and 294 g.

(ii) 16 brand *A* peaches are randomly selected. Someone claims that the probability that the mean weight of the selected peaches lies between 274 g and 294 g is greater than the probability obtained in (a)(i). Do you agree? Explain your answer.

(4 marks)

(b) The wholesaler also buys a large batch of brand *B* peaches. The weights of this batch of peaches follow a normal distribution with a mean of μ g and a standard deviation of σ g. In this batch, 33% of the weights of peaches are more than 281.2 g and 12.3% of the weights of peaches are less than 273.2 g.

(i) Find μ and σ.

(ii) A peach is said to be *excellent* if its weight lies between 274 g and 294 g. The batch, either of brand *A* or brand *B*, with a higher percentage of *excellent* peaches is now selected. 8 peaches are then randomly selected one by one from that batch. Given that more than 3 *excellent* peaches are selected, find the probability that the 7th peach selected is the 5th *excellent* peach.

(8 marks)

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11. In a research, the rate of change of profit (in million dollars per month) made by company A is modelled by

$$A(t) = 3t \ln (t^2 + 2t + 10),$$

where t ($0 \le t \le 12$) is the number of months elapsed since the research starts. Denote the total profit made by company A in the first 4 months by α million dollars. Let α_1 be the estimate of α by using the trapezoidal rule with 4 sub-intervals.

(a) (i) Find α_1.

(ii) Is α_1 an over-estimate or an under-estimate? Explain your answer.

(6 marks)

(b) In the same research, the rate of change of profit (in million dollars per month) made by company B is modelled by

$$B(t) = \frac{16t + 3}{\sqrt{t + 1}},$$

where t ($0 \le t \le 12$) is the number of months elapsed since the research starts. Denote the total profit made by company B in the first 4 months by β million dollars.

(i) Find β.

(ii) A researcher claims that the difference between the total profits made by the two companies in the first 4 months since the research starts exceeds 1 million dollars. Do you agree? Explain your answer.

(6 marks)

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12. The number of messages N (in thousand) sent to a social media account of an internet celebrity can be modelled by

$$N = \frac{240}{h^{4+kt} + 5},$$

where h and k are constants and $t(t \geq 0)$ is the number of days elapsed since the break of a scandal involving the internet celebrity.

(a) Express $\ln\left(\frac{240 - 5N}{N}\right)$ as a linear function of t. (2 marks)

(b) It is given that the graph of the linear function obtained in **(a)** passes through the point $(5, \ln 27)$ and the intercept on the horizontal axis of the graph is 20.

(i) Find h and k.

(ii) Find $\frac{dN}{dt}$ and $\frac{d^2N}{dt^2}$.

(iii) Estimate the number of messages sent to the social media account of the internet celebrity after a very long time. Hence, or otherwise, prove that the number of messages sent to the social media account does not exceed 50 000 since the break of the scandal.

(iv) Find the number of messages sent to the social media account of the internet celebrity when $\frac{dN}{dt}$ attains its greatest value.

(11 marks)

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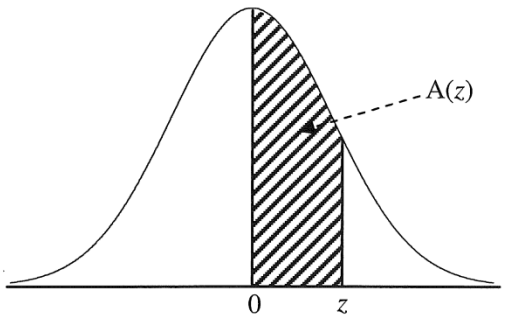
Answers written in the margins will not be marked.

End of Paper

Table: Area under the Standard Normal Curve

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4992	.4993	.4993
3.2	.4993	.4993	.4994	.4994	.4994	.4994	.4994	.4995	.4995	.4995
3.3	.4995	.4995	.4995	.4996	.4996	.4996	.4996	.4996	.4996	.4997
3.4	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4997	.4998
3.5	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998	.4998

Note : An entry in the table is the proportion of the area under the entire curve which is between $z = 0$ and a positive value of z. Areas for negative values of z are obtained by symmetry.



$$A(z) = \int_0^z \frac{1}{\sqrt{2\pi}} e^{\frac{-x^2}{2}} dx$$

2425 S6 Final Exam Marking Scheme

1. (a) $0.1 + a + b + 0.15 = 1$

$a = 0.75 - b$ ..... (1)

$\mathrm{E}(X) = 0(0.1) + 1(a) + 2(b) + 3(0.15)$

$= a + 2b + 0.45$

1M

$\mathrm{Var}(X) = 0.75$

$0^2(0.1) + 1^2(a) + 2^2(b) + 3^2(0.15) - (a + 2b + 0.45)^2 = 0.75$

$a + 4b - (a + 2b + 0.45)^2 = -0.6$ ..... (2) 1M

By substituting (1) into (2), we have

$0.75 - b + 4b - (0.75 - b + 2b + 0.45)^2 = -0.6$

$0.75 + 3b - (b + 1.2)^2 = -0.6$ 1A ----- for both correct

$0.75 + 3b - b^2 - 2.4b - 1.44 = -0.6$

$b^2 - 0.6b + 0.09 = 0$

$b = \underline{\underline{0.3}}$

$a = 0.75 - 0.3$

$= \underline{\underline{0.45}}$ 1M

(b) $P(G \mid F) = \dfrac{P(F \cap G)}{P(F)}$ 1A

$= \dfrac{0.45 + 0.3}{0.45 + 0.3 + 0.15}$

$= \underline{\underline{\dfrac{5}{6}}}$

(6)

2. (a) $\mathrm{P}(A \mid B') - \mathrm{P}(B' \mid A) = \dfrac{5}{13}$ 1M

$\dfrac{\mathrm{P}(A \cap B')}{\mathrm{P}(B')} - \dfrac{\mathrm{P}(A \cap B')}{\mathrm{P}(A)} = \dfrac{5}{13}$

$\dfrac{\mathrm{P}(A \cap B')}{\mathrm{P}(B')} - \dfrac{\mathrm{P}(A \cap B')}{2\,\mathrm{P}(B')} = \dfrac{5}{13}$

$\dfrac{\mathrm{P}(A \cap B')}{2\,\mathrm{P}(B')} = \dfrac{5}{13}$

$\dfrac{\mathrm{P}(A \cap B')}{\mathrm{P}(B')} = \dfrac{10}{13}$

$\mathrm{P}(A \mid B') = \dfrac{10}{13}$

		1A
(b)	$\frac{P(A \cap B')}{P(B')} = \frac{10}{13}$	
	$P(B') = 1.3\ P(A \cap B')$	
	$P(A) = 2\ P(B')$	
	$= 2.6\ P(A \cap B')$	
	$P(A \cup B') = P(A) + P(B') - P(A \cap B')$	
	$0.58 = 2.6\ P(A \cap B') + 1.3\ P(A \cap B') - P(A \cap B')$	1M
	$0.58 = 2.9\ P(A \cap B')$	
	$P(A \cap B') = 0.2$	
	$P(A) = 2.6 \times 0.2$	
	$= 0.52$	
	$P(A \cap B) = P(A) - P(A \cap B')$	1A
	$= 0.52 - 0.2$	
	$= 0.32$	
		1A
(c)	$P(B') = 1.3 \times 0.2$	
	$= 0.26$	
	$P(B) = 1 - P(B')$	
	$= 1 - 0.26$	
	$= 0.74$	
	$P(A) \times P(B) = 0.52 \times 0.74 = 0.384\ 8$	
	$P(A \cap B) = 0.32$	
	$\therefore$ $P(A) \times P(B) \neq P(A \cap B)$	1M
	$\therefore$ A and B are not independent.	1A
		----------(7)

3. **(a)** The required probability

$= 1-(1-0.03)^5$ 1M

$\approx 0.141\,265\,974$

$= \underline{\underline{0.1413}}$ (cor. to 4 d.p.) 1A

(b) The expected number of accepted batches

$\approx \dfrac{1}{0.141\,265\,974} - 1$ 1M

$\approx 6.078\,845\,454$

$= \underline{\underline{6.0788}}$ (cor. to 4 d.p.) 1A

(c) The required probability 1M

$\approx (1-0.141\,265\,974)^{15} + C_{14}^{15}(1-0.141\,265\,974)^{14}(0.141\,265\,974)$
$+ C_{13}^{15}(1-0.141\,265\,974)^{13}(0.141\,265\,974)^2$

$\approx 0.642\,459\,011$

$= \underline{\underline{0.6425}}$ (cor. to 4 d.p.) 1A

(6)

4. (a) $E(X) = 10 \times 60$

$= 600$ 1A

$\text{Var}(X) = E(X)$

$= 600$ 1A

(b) By central limit theorem, $\bar{X} \sim N\left(600, \dfrac{600}{n}\right)$ approximately. 1M

$$P(590 \le \bar{X} \le 610) > 0.994$$

$$P\left(\frac{590-600}{\sqrt{\frac{600}{n}}} \le Z \le \frac{610-600}{\sqrt{\frac{600}{n}}}\right) > 0.994$$ 1M

$$P\left(-10\sqrt{\frac{n}{600}} \le Z \le 10\sqrt{\frac{n}{600}}\right) > 0.994$$

$$P\left(0 \le Z \le 10\sqrt{\frac{n}{600}}\right) > 0.497$$

$$10\sqrt{\frac{n}{600}} > 2.75$$

$$n > 45.375$$ 1A

$\therefore$ The least value of n is 46.

----------(5)

5. (a) Let $u = x^m$. Then $du = mx^{m-1}dx$. — 1M

$$\int f(x)\,dx = \int e^{x^m} x^{m-1} dx$$

$$= \int \frac{1}{m} e^u du$$

$$= \frac{1}{m} e^u + C$$

$$= \frac{1}{m} e^{x^m} + C$$ — 1M

1A

Let $u = e^{x^m}$. Then $du = e^{x^m}(mx^{m-1})dx$. — 1M

$$\int f(x)\,dx = \int e^{x^m} x^{m-1} dx$$

$$= \int \frac{1}{m} du$$

$$= \frac{1}{m} u + C$$

$$= \frac{1}{m} e^{x^m} + C$$ — 1M

1A

(b) $\int_1^8 e^{x^k} x^{k-1} dx = \frac{e}{k}(e-1)$ — 1M

$$\left[\frac{1}{k} e^{x^k}\right]_1^8 = \frac{e}{k}(e-1)$$ — 1M

$$\frac{1}{k} e^{8^k} - \frac{1}{k} e = \frac{1}{k} e^2 - \frac{1}{k} e$$

$$\frac{1}{k} e^{8^k} = \frac{1}{k} e^2$$

$$e^{8^k} = e^2$$

$$8^k = 2$$

$$2^{3k} = 2$$

$$3k = 1$$

$$k = \frac{1}{3}$$

1A

----------(6)

6. (a) $e^{-ax} = 1+(-ax)+\frac{(-ax)^2}{2!}+\frac{(-ax)^3}{3!}+\cdots$

$= 1-ax+\frac{a^2x^2}{2}-\frac{a^3x^3}{6}+\cdots$ 1A

(b)(i) $y = (1 + ax)^5e^{-ax}$

$=(1+5ax+10a^2x^2+10a^3x^3+\cdots)\left(1-ax+\frac{a^2x^2}{2}-\frac{a^3x^3}{6}+\cdots\right)$ 1M

$\because$ Coefficient of $x^3 = 63$

$\therefore \quad (1)\left(-\frac{a^3}{6}\right)+(5a)\left(\frac{a^2}{2}\right)+(10a^2)(-a)+(10a^3)(1) = 63$

$-\frac{a^3}{6}+\frac{5a^3}{2}-10a^3+10a^3 = 63$

$\frac{7a^3}{3} = 63$

$a^3 = 27$

$a = 3$ 1A

(ii) $\frac{dy}{dx} = 5(3)(1 + 3x)^4e^{-3x} + (–3)(1 + 3x)^5e^{-3x}$ 1M

$= 3(1 + 3x)^4(5 – 1 – 3x)e^{-3x}$

$= 3(1 + 3x)^4(4 – 3x)e^{-3x}$

When $\frac{dy}{dx} = 0$,

$3(1 + 3x)^4(4 – 3x)e^{-3x} = 0$

$(1 + 3x)^4(4 – 3x) = 0$

$x = -\frac{1}{3}$ *or* $\frac{4}{3}$ 1M

x	$x<-\frac{1}{3}$	$x=-\frac{1}{3}$	$-\frac{1}{3}<x<\frac{4}{3}$	$x=\frac{4}{3}$	$x>\frac{4}{3}$
$\frac{dy}{dx}$	+	0	+	0	–

1M

The graph has only one turning point at $x = \frac{4}{3}$.

$\therefore$ The claim is agreed. 1A

----------(7)

7. (a) $\dfrac{dy}{dx} = \dfrac{(2x^2+4)^{\frac{1}{2}}(3x^2-4) - (x^3-4x)\left[\dfrac{1}{2}(2x^2+4)^{-\frac{1}{2}}(4x)\right]}{2x^2+4}$ 1M

$= \dfrac{(2x^2+4)^{-\frac{1}{2}}[(2x^2+4)(3x^2-4) - 2x(x^3-4x)]}{2x^2+4}$

$= \dfrac{2(3x^4-4x^2+6x^2-8-x^4+4x^2)}{(2x^2+4)^{\frac{3}{2}}}$

$= \dfrac{2(2x^4+6x^2-8)}{(2x^2+4)^{\frac{3}{2}}}$

$= \dfrac{4(x^4+3x^2-4)}{(2x^2+4)^{\frac{3}{2}}}$ 1A

(b) When $\dfrac{dy}{dx} = 0$,

$x^4+3x^2-4=0$ 1M

$(x^2-1)(x^2+4)=0$

$x^2=1$ or $x^2=-4$ (rejected)

$x=\pm 1$

x	$-2<x<-1$	$x=-1$	$-1<x<1$	$x=1$	$1<x<4$
y	⟋	$\dfrac{\sqrt{6}}{2}$	⟋	$-\dfrac{\sqrt{6}}{2}$	⟋
$\dfrac{dy}{dx}$	+	0	−	0	+

1M

When $x=-2$, $y=\dfrac{(-2)^3-4(-2)}{\sqrt{2(-2)^2+4}}=0$

When $x=4$, $y=\dfrac{4^3-4(4)}{\sqrt{2(4)^2+4}}=8$

$\therefore$ y attains its greatest value at $x=4$, and y attains its least value at $x=1$.

$\therefore$ The greatest value of y is 8, and the least value of y is $-\dfrac{\sqrt{6}}{2}$. 1A + 1A

(6)

8. (a) $\dfrac{d}{dx}[(2x^2+3x)e^{-2x}] = (2x^2+3x)[e^{-2x}(-2)] + e^{-2x}(4x+3)$ 1M

$= e^{-2x}[-2(2x^2+3x)+(4x+3)]$

$= e^{-2x}(-4x^2-2x+3)$

$\therefore \quad (2x^2+3x)e^{-2x} = \int e^{-2x}(-4x^2-2x+3)dx$

$= -2\int (2x^2+x)e^{-2x}dx + 3\int e^{-2x}dx$

$\int (2x^2+x)e^{-2x}dx = -\dfrac{1}{2}(2x^2+3x)e^{-2x} + \dfrac{3}{2}\int e^{-2x}dx$ 1M

$= -\dfrac{1}{2}(2x^2+3x)e^{-2x} - \dfrac{3}{4}e^{-2x} + \text{Constant}$

$= -\dfrac{1}{4}(4x^2+6x+3)e^{-2x} + \text{Constant}$ 1A

(b) $f(x) = (4x^2+2x+1)e^{-2x}$

$f'(x) = (4x^2+2x+1)[e^{-2x}(-2)] + e^{-2x}(8x+2)$

$= e^{-2x}[-2(4x^2+2x+1)+(8x+2)]$

$= e^{-2x}(-8x^2+4x)$

$= -4x(2x-1)e^{-2x}$ 1A

When $f'(x) = 0$,

$-4x(2x-1)e^{-2x} = 0$

$x(2x-1) = 0$

$x = 0 \quad \text{or} \quad \dfrac{1}{2}$

$\because \quad \alpha > \beta$

$\therefore \quad \alpha = \dfrac{1}{2},\ \beta = 0$ 1A

$\int_{\beta}^{\alpha} f(x)dx = \int_0^{\frac{1}{2}} (4x^2+2x+1)e^{-2x}dx$

$= 2\int_0^{\frac{1}{2}} (2x^2+x)e^{-2x}dx + \int_0^{\frac{1}{2}} e^{-2x}dx$

$= 2\left[-\dfrac{1}{4}(4x^2+6x+3)e^{-2x}\right]_0^{\frac{1}{2}} + \left[-\dfrac{1}{2}e^{-2x}\right]_0^{\frac{1}{2}}$ 1M

$= 2\left\{-\dfrac{1}{4}\left[4\left(\dfrac{1}{2}\right)^2 + 6\left(\dfrac{1}{2}\right)+3\right]e^{-2\left(\frac{1}{2}\right)} - \left(-\dfrac{1}{4}\right)[4(0)^2+6(0)+3]e^{-2(0)}\right\}$

$+ \left\{\left[-\dfrac{1}{2}e^{-2\left(\frac{1}{2}\right)}\right] - \left[-\dfrac{1}{2}e^{-2(0)}\right]\right\}$

$= 2\left(-\dfrac{7}{4}e^{-1}+\dfrac{3}{4}\right) + \left(-\dfrac{1}{2}e^{-1}+\dfrac{1}{2}\right)$

$= 2-4e^{-1}$ 1A

(7)

9. (a) Let X be the number of requests received by the company for the repair service in a day, then $X \sim \text{Po}(2.1)$.

Required probability $= P(X < 6)$

$$= e^{-2.1} + \frac{e^{-2.1}(2.1)^1}{1!} + \frac{e^{-2.1}(2.1)^2}{2!} + \frac{e^{-2.1}(2.1)^3}{3!} + \frac{e^{-2.1}(2.1)^4}{4!} + \frac{e^{-2.1}(2.1)^5}{5!} \quad \text{1M}$$

$$\approx 0.979\,550\,919$$

$$= 0.979\,6 \text{ (cor. to 4 d.p.)} \quad \text{1A}$$

(2)

(b) $P(X > 2) = 1 - e^{-2.1} - \frac{e^{-2.1}(2.1)^1}{1!} - \frac{e^{-2.1}(2.1)^2}{2!}$

$$\approx 0.350\,368\,648$$

Let Y be the number of days with more than 2 repair service requests out of n days, then $Y \sim \text{B}(n, p)$ where $p = P(X > 2)$.

$$P(Y \geq 1) > 0.99$$

$$1 - (0.649\,631\,351)^n > 0.99 \quad \text{1M}$$

$$(0.649\,631\,351)^n < 0.01$$

$$n \log 0.649\,631\,351 < \log 0.01$$

$$n > 10.68 \text{ (cor. to 2 d.p.)}$$

$\therefore$ The least integral value of n is 11. 1A

(2)

(c) (i) Let W_i be the number of repair service requests which are charged \$1 000 or above out of i repair service requests where i is a positive integer, then $W_i \sim \text{B}(i, 0.6)$.

Required probability $= P(X = 5) \times P(W_5 \leq 1)$

$$= \frac{e^{-2.1}(2.1)^5}{5!} \times [(0.4)^5 + C_1^5(0.6)(0.4)^4] \quad \text{1M}$$

$$\approx 0.003\,627\,569\,134$$

$$= 0.003\,6 \text{ (cor. to 4 d.p.)} \quad \text{1A}$$

(ii) Required probability

$$= P(X \leq 1) + P(X = 2) \times P(W_2 \leq 1) + P(X = 3) \times P(W_3 \leq 1)$$
$$+ P(X = 4) \times P(W_4 \leq 1) + P(X = 5) \times P(W_5 \leq 1)$$

$$\approx e^{-2.1} + \frac{e^{-2.1}(2.1)^1}{1!} + \frac{e^{-2.1}(2.1)^2}{2!} \times [(0.4)^2 + C_1^2(0.6)(0.4)] + \frac{e^{-2.1}(2.1)^3}{3!} \times [(0.4)^3$$
$$+ C_1^3(0.6)(0.4)^2] + \frac{e^{-2.1}(2.1)^4}{4!} \times [(0.4)^4 + C_1^4(0.6)(0.4)^3] + 0.003\,627\,569\,134 \quad \text{1M + 1M}$$

$$\approx 0.640\,367\,256$$

$$= 0.640\,4 \text{ (cor. to 4 d.p.)} \quad \text{1A}$$

(iii) Let A be the event that a day is considered as a bad day, and B be the event that exactly 1 repair service request is charged \$1 000 or above on a day.

$P(B \mid A)$

$= [P(X = 1) \times P(W_1 = 1) + P(X = 2) \times P(W_2 = 1) + P(X = 3) \times P(W_3 = 1)$

$+ P(X = 4) \times P(W_4 = 1) + P(X = 5) \times P(W_5 = 1)] \div P(A)$

$\approx \left[\frac{e^{-2.1}(2.1)^1}{1!} \times 0.6 + \frac{e^{-2.1}(2.1)^2}{2!} \times C_1^2 (0.6)(0.4) + \frac{e^{-2.1}(2.1)^3}{3!} \times C_1^3 (0.6)(0.4)^2 \right.$

$\left. + \frac{e^{-2.1}(2.1)^4}{4!} \times C_1^4 (0.6)(0.4)^3 + \frac{e^{-2.1}(2.1)^5}{5!} \times C_1^5 (0.6)(0.4)^4 \right] \div 0.640\,367\,256$ 1M + 1M

$\approx 0.557\,150\,563$

Let U be the number of bad days on which exactly 1 repair service request is charged \$1 000 or above out of 7 bad days, then $U \sim \text{B}(7, q)$ where $q = P(B \mid A)$.

Required probability

$= P(U \geq 6)$

$\approx C_6^7 (0.557\,150\,563)^6 (0.442\,849\,436) + (0.557\,150\,563)^7$ 1M

$\approx 0.109\,388\,553$

$= 0.109\,4$ (cor. to 4 d.p.) 1A

(9)

10. (a)(i) Let X g be the weight of a brand A peach.

Then $X \sim \text{N}(284, 20^2)$.

The required probability

$= \text{P}(274 < X < 294)$

$= \text{P}\left(\frac{274 - 284}{20} < Z < \frac{294 - 284}{20} \right)$ 1M

$= \text{P}(-0.5 < Z < 0.5)$

$= 0.191\,5 + 0.191\,5$

$= 0.383$ 1A

(ii) Let $\overline{X}$ g be the mean weight of 16 brand A peaches.

Then $\overline{X} \sim \text{N}\left(284, \frac{20^2}{16} \right)$.

$\text{P}(274 < \overline{X} < 294)$

$= \text{P}\left(\frac{274 - 284}{\frac{20}{\sqrt{16}}} < Z < \frac{294 - 284}{\frac{20}{\sqrt{16}}} \right)$ 1M

$= \text{P}(-2 < Z < 2)$

$= 0.477\,2 + 0.477\,2$

$= 0.954\,4$

> 0.383 1A

$\therefore$ The claim is agreed.

----------(4)

(b)(i) Let Y g be the weight of a brand B peach.

Then $Y \sim N(\mu, \sigma^2)$.

$$P(Y > 281.2) = 0.33$$

$$P\left(Z > \frac{281.2-\mu}{\sigma}\right) = 0.33$$

$$P\left(0 \le Z \le \frac{281.2-\mu}{\sigma}\right) = 0.5 - 0.33$$

$$P\left(0 \le Z \le \frac{281.2-\mu}{\sigma}\right) = 0.17$$

From the standard normal table,

$$\frac{281.2-\mu}{\sigma} = 0.44$$

$$281.2 - \mu = 0.44\sigma \ldots\ldots\ldots\ldots\ldots (1)$$

$$P(Y < 273.2) = 0.123$$

$$P\left(Z < \frac{273.2-\mu}{\sigma}\right) = 0.123$$

$$P\left(0 \le Z \le -\frac{273.2-\mu}{\sigma}\right) = 0.5 - 0.123$$

$$P\left(0 \le Z \le -\frac{273.2-\mu}{\sigma}\right) = 0.377$$

From the standard normal table,

$$-\frac{273.2-\mu}{\sigma} = 1.16$$

$$273.2 - \mu = -1.16\sigma \ldots\ldots\ldots\ldots (2)$$

(1) – (2): $8 = 1.6\sigma$ 1A

$\sigma = 5$

Substitute $\sigma = 5$ into (1). 1A

$281.2 - \mu = 0.44(5)$

$\mu = 279$

1M for either one

(ii) $P(274 < Y < 294)$

$$= P\left(\frac{274-279}{5} < Z < \frac{294-279}{5}\right)$$ 1M

$= P(-1 < Z < 3)$

$= 0.341\,3 + 0.498\,7$

$= 0.84$

> 0.383 1M

$\therefore$ The batch of brand B has a higher percentage of excellent peaches.

The required probability

$$= \frac{C_4^6(0.84)^4(1-0.84)^2(0.84)}{C_4^8(1-0.84)^4(0.84)^4 + C_5^8(1-0.84)^3(0.84)^5 + C_6^8(1-0.84)^2(0.84)^6 + C_7^8(1-0.84)(0.84)^7 + 0.84^8}$$ 1M+1M

$= 0.161\,2$, *cor. to 4 d.p.* 1A

----------(8)

11. (a)(i) $\alpha = \int_0^4 3t \ln(t^2 + 2t + 10)\, dt$

Length of each sub-interval = $\frac{4-0}{4} = 1$

t	0	1	2
$A(t)$	0	7.694 848 072	17.342 230 55

t	3	4
$A(t)$	28.969 882 42	42.316 326 3

$\alpha_1 \approx \frac{1}{2}[0 + 2(7.694\,848\,072 + 17.342\,230\,55 + 28.969\,882\,42) + 42.316\,326\,3]$

$\approx 75.165\,124\,19$

$= 75.165\,1$, *cor. to 4 d.p.*

1M

1A

(ii) $A'(t) = 3 \ln (t^2 + 2t + 10) + 3t\left(\frac{1}{t^2 + 2t + 10}\right)(2t + 2)$ 1M

$= 3 \ln (t^2 + 2t + 10) + \frac{6t^2 + 6t}{t^2 + 2t + 10}$

$A''(t) = 3\left(\frac{1}{t^2 + 2t + 10}\right)(2t + 2) + \frac{(t^2 + 2t + 10)(12t + 6) - (6t^2 + 6t)(2t + 2)}{(t^2 + 2t + 10)^2}$

1M

$= \frac{(6t + 6)(t^2 + 2t + 10)}{(t^2 + 2t + 10)^2} + \frac{6t^2 + 120t + 60}{(t^2 + 2t + 10)^2}$

$= \frac{6t^3 + 18t^2 + 72t + 60}{(t^2 + 2t + 10)^2} + \frac{6t^2 + 120t + 60}{(t^2 + 2t + 10)^2}$ 1A

$= \frac{6t^3 + 24t^2 + 192t + 120}{(t^2 + 2t + 10)^2}$

For $0 \le t \le 4$, $A''(t) > 0$.

$\therefore$ α_1 is an over-estimate.

1A

----------(6)

(b)(i) $\beta = \int_0^4 \frac{16t+3}{\sqrt{t+1}}\,dt$

Let $u = t + 1$. Then $du = dt$.

When $t = 0$, $u = 0 + 1 = 1$;

when $t = 4$, $u = 4 + 1 = 5$.

1M

$$\beta = \int_0^4 \frac{16t+3}{\sqrt{t+1}}\,dt$$

$$= \int_1^5 \frac{16(u-1)+3}{\sqrt{u}}\,du$$

$$= \int_1^5 \left(16u^{\frac{1}{2}} - 13u^{-\frac{1}{2}}\right)du$$

$$= \left[\frac{32}{3}u^{\frac{3}{2}} - 26u^{\frac{1}{2}}\right]_1^5$$ 1M

$$= \left[\frac{32}{3}(5)^{\frac{3}{2}} - 26(5)^{\frac{1}{2}}\right] - \left(\frac{32}{3} - 26\right)$$ 1M

$\approx 76.452\ 524\ 72$

$= 76.452\ 5$, *cor. to 4 d.p.*

1A

(ii) From (a)(ii), α_1 is an over-estimate of α.

$\therefore \quad \alpha_1 > \alpha$

$\beta - \alpha > \beta - \alpha_1$ 1M

$\approx 76.452\ 524\ 72 - 75.165\ 124\ 19$

$= 1.287\ 400\ 53$

> 1

$\therefore$ The difference between the total profits made by the two companies in the first 4 months since the research starts exceeds 1 million dollars.

$\therefore$ The claim is agreed.

1A

----------(6)

12. (a) $N = \dfrac{240}{h^{4+kt} + 5}$

$Nh^{4+kt} + 5N = 240$

$\dfrac{240 - 5N}{N} = h^{4+kt}$ 1M

$\ln\left(\dfrac{240 - 5N}{N}\right) = \ln h^{4+kt}$

$\therefore \quad \ln\left(\dfrac{240 - 5N}{N}\right) = 4\ln h + (k \ln h)t$ 1A

(2)

(b) (i) $\because$ The intercept on the horizontal axis of the graph is 20.

$\therefore \quad 0 = 4\ln h + (k \ln h)(20)$

$20k \ln h = -4 \ln h$

$k = -0.2$ 1A

$\because$ The graph passes through (5, ln 27).

$\therefore \quad \ln 27 = 4\ln h + (-0.2 \ln h)(5)$

$\ln 27 = 3 \ln h$

$\ln h = \dfrac{1}{3} \ln 27$

$= \ln 3$

$\therefore \quad h = 3$ 1A

(ii) $N = \dfrac{240}{3^{4-0.2t} + 5}$

$\dfrac{dN}{dt} = 240[-(3^{4-0.2t} + 5)^{-2}][3^{4-0.2t}(\ln 3)(-0.2)]$ 1M

$= \dfrac{3\,888(\ln 3)(3^{-0.2t})}{(3^{4-0.2t} + 5)^2}$ 1A

$\dfrac{d^2N}{dt^2} = 3\,888(\ln 3)\left\{\dfrac{(3^{4-0.2t} + 5)^2(3^{-0.2t} \ln 3)(-0.2)}{(3^{4-0.2t} + 5)^4}\right.$

$\left. - \dfrac{(3^{-0.2t})[2(3^{4-0.2t} + 5)(3^{4-0.2t} \ln 3)(-0.2)]}{(3^{4-0.2t} + 5)^4}\right\}$ 1M

$= 3\,888(\ln 3)(3^{4-0.2t} + 5)(3^{-0.2t} \ln 3)(-0.2) \times \dfrac{(3^{4-0.2t} + 5) - 2(3^{4-0.2t})}{(3^{4-0.2t} + 5)^4}$

$= -\dfrac{777.6(\ln 3)^2(3^{-0.2t})(5 - 3^{4-0.2t})}{(3^{4-0.2t} + 5)^3}$ 1A

(iii) $\lim\limits_{t\to\infty} N = \lim\limits_{t\to\infty} \dfrac{240}{3^{4-0.2t}+5}$

$= \dfrac{240}{0+5}$

$= 48$

$\therefore$ Estimated number of messages sent to the social media account of the internet celebrity after a very long time is 48 000. 1A

$\because$ $\dfrac{dN}{dt} > 0$ for $t \geq 0$

$\therefore$ N increases for $t \geq 0$.

$\because$ N increases for $t \geq 0$ and $\lim\limits_{t\to\infty} N = 48 < 50$

$\therefore$ The number of messages sent to the social media account of the internet celebrity does not exceed 50 000 since the break of the scandal. 1

(iv) When $\dfrac{d^2N}{dt^2} = 0$,

$5 - 3^{4-0.2t} = 0$ 1M

$3^{-0.2t} = \dfrac{5}{3^4}$

$-(0.2\ln 3)t = \ln 5 - 4\ln 3$

$t = 20 - \dfrac{5\ln 5}{\ln 3}$

t	$0 < t < 20 - \dfrac{5\ln 5}{\ln 3}$	$t = 20 - \dfrac{5\ln 5}{\ln 3}$	$t > 20 - \dfrac{5\ln 5}{\ln 3}$
$\dfrac{d^2N}{dt^2}$	+	0	–

1M

$\therefore$ $\dfrac{dN}{dt}$ attains its greatest value at $t = 20 - \dfrac{5\ln 5}{\ln 3}$.

When $t = 20 - \dfrac{5\ln 5}{\ln 3}$,

$N = \dfrac{240}{3^{4-0.2\left(20-\frac{5\ln 5}{\ln 3}\right)}+5}$

$= 24$

$\therefore$ The number of messages sent to the social media account of the internet celebrity is 24 000 when $\dfrac{dN}{dt}$ attains its greatest value. 1A

(11)